The Induction Principle — problem 8.12

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We are given N lines ($N > 1$) in a plane, no two of which are parallel and no three of which have a point in common. Prove that it is possible to assign a non-zero integer of absolute value not exceeding N to each region of the plane determined by these lines, such that the sum of the integers on either side of any of the given lines is equal to 0 (TT 1989).